

C++ Constructors

A constructor is a special member function for automatic initialization of an object. A constructor is called automatically (implicit) when an object of class is created.

Rules for declaring constructor:

- I. In C++, a constructor has the same name as that of the class.
- II. It does not have a return type (not even void).
- III. It cannot be declared CONST or VOLATILE.
- IV. It may not be STATIC or VIRTUAL
- V. It should have PUBLIC access.
- VI. It takes argument and therefore it can be overload.

The general syntax of constructor:

Class Test

```
{  
    private :  
    public : Test();  
    protected:  
};
```

Test : : Test()

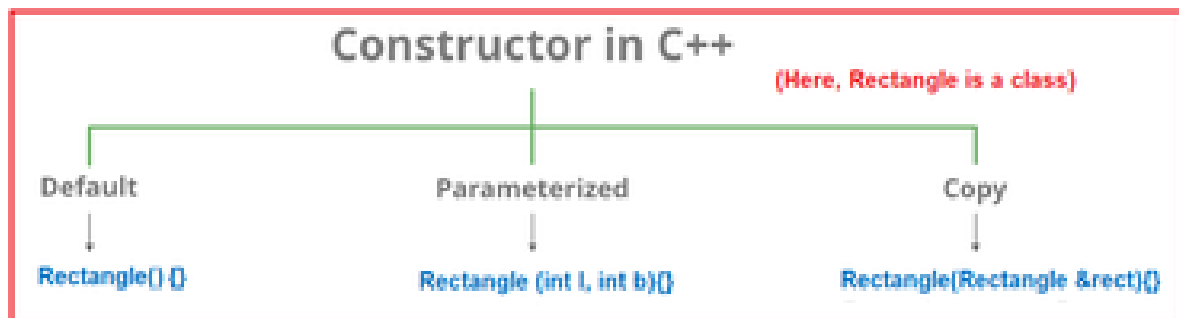
```
{  
    Cout<<"Default Constructor";  
}
```

main()

```
{  
    clrscr();  
    Test o;  
    getch();  
}
```

Output : Default Constructor

C++ offers three types of constructors:



Default Constructor: A constructor with no parameters is known as a default constructor. Initializes object attributes with default values. In the example above, **Rectangle ()** is a default constructor.

Parameterized Constructor: In C++, a constructor with parameters is known as a parameterized constructor. This is the preferred method to initialize member data. In the example above, **Rectangle (int l, int b)** is a parameterized constructor.

Copy Constructor: The copy constructor in C++ is used to copy data of one object to another. It is based on **REFERENCE (&)**. In the example above, **Rectangle (Rectangle &p)** is a copy constructor.

Question:

- C++ Program to Add Two Numbers using default constructor.
- C++ Program to Add Two Numbers using parameterized constructor.
- C++ Program to find factorial of a number using default constructor.
- C++ Program to find sum of first N natural number, N must be taken by the user.
- C++ program to print all prime numbers from 1 to N.
- C++ program to print all even and odd numbers from 1 to N.
- C++ program to print all Armstrong numbers from 1 to N.
- C++ program to print square, cube and square root of all numbers from 1 to N.
- C++ program to print all leap years from 1 to N using parameterized constructor.

Constructor Overloading: Constructors can be overloaded in a similar way as function overloading. C++ allows us to overload constructors, which means we can define multiple constructors within the same class, as long as each has a different list of arguments.

Example : Find area of Rectangle Using Constructor Overloading.

```
class Rect
{
    private:
        int area;
    public:
        Rect(); // Default Constructor
        Rect (int , int); // Parameterized Constructor
        Void disp();
};

Rect :: Rect ()
{
    area = 0;
}

Rect :: Rect ( int a, int b)
{
    area = a * b;
}

void Rect :: disp ()
{
    cout << "The area is: " << area << endl;
}

main()
{
    Rect r1;
    Rect r2(2, 6);
    r1.disp();
    r2.disp();
}
```

Question: Find area of Circle, Rectangle, Triangle using Constructor Overloading.

Destructors in C++

Destructor is a special member function that is executed automatically (implicit) whenever an object is going to be destroyed.

Destructors are usually used to deallocate memory and do other cleanup for a class object and its class members when the object is destroyed.

Rules for declaring Destructor:

- I. A destructor has the same name as the class and is preceded by a tilde (~).
- II. It does not have a return type (not even void).
- III. It cannot be declared CONST or VOLATILE.
- IV. It may not be STATIC or VIRTUAL
- V. It should have PUBLIC access.
- VI. It takes no argument and therefore it cannot be overload.

The general syntax of destructor:

Class Test

```
{
    private :
    public : Test();    // Default Constructor
             ~Test();  // Destructor
    protected:
```

```
};
```

```
Test : : Test()
```

```
{
    Cout<<"Default Constructor"<<endl;
}
```

```
Test : : ~Test()
```

```
{
    Cout<<"Memory Released";
}
```

```
main()
```

```
{
    clrscr();
    Test o;
    getch();
}
```

Output : Default Constructor
Memory Released

Difference Between Constructor and Destructor in C++

SNO	Constructor	Destructor
1	A constructor has the same name as that of the class	A destructor has the same name as the class and is preceded by a tilde (~).
2	It should have PUBLIC or PROTECTED access	It should have PUBLIC access
3	A constructor is called automatically (implicit) when an object of class is created.	A destructor executed automatically (implicit) whenever an object is going to be destroyed.
4	It receives arguments.	It does not receive any argument.
5	It can be overloaded.	It cannot be overloaded.
6	In a class, there can be multiple constructors.	While in a class, there is always a single destructor.
7	They are often called in successive order.	They are often called in reverse order of constructor.
8	Constructor is used to allocate the memory to an instance or object.	While it is used to deallocate the memory of an object of a class.